KS3 DIGITAL TECHNOLOGY

DEEPFAKES AND MISINFORMATION

How fake images, videos, voices and stories are made and spread, how professional fact-checkers catch them, and what you can do to stop them.

LESSONS

1 Fake news? It's complicated
2 How deepfakes are made
3 Fact-checker's toolkit
4 Think like a fact-checker
5 Why lies travel fast
6 Fighting back

LESSON 1 OF 6

FAKE NEWS? IT'S COMPLICATED

Not all false information is the same.

STARTER · 5 MINUTES

DO NOW

Your gran shares a post saying "Drinking hot water cures colds!" Is she lying? What's the difference between being wrong and lying?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain the difference between misinformation, disinformation and malinformation

SUCCESS CRITERIA: I CAN...

- ✓ I can define misinformation, disinformation and malinformation
- ✓ I can sort real examples into the three types
- ✓ I can explain why intent matters

THREE KINDS OF "FAKE"

- Misinformation: false, but shared by someone who believes it
- Disinformation: false, and shared on purpose to deceive
- Malinformation: true, but shared to cause harm
- The difference is the intent: what the sharer is trying to do

KEY WORDS

MISINFORMATION	False information shared by someone who believes it is true.
DISINFORMATION	False information created or shared on purpose to deceive.
MALINFORMATION	True information shared on purpose to cause harm.

ACTIVITY · 15 MINUTES · PAIRS

SORT THE STORIES

1. Read the eight short scenarios on the board or card set.
2. Decide whether each is misinformation, disinformation, malinformation, or true and harmless.
3. For each one, write the clue that helped you decide.

In your workbook, answer:

? Which scenario was hardest to sort, and why?

? Explain why the person's intent matters when we decide how to respond.

Your workbook: Lesson 1 · tick each step off as you go

WHY PEOPLE MAKE DISINFORMATION

- Money: clicks, adverts and scams
- Power: changing how people vote or think
- Mischief or revenge: embarrassing someone
- Chaos: making people distrust everything

WHY IT MATTERS

- False health information can make people ill
- False stories about groups of people can stir up hate
- Elections depend on voters having accurate information
- If we can't agree on facts, we can't solve problems together

QUICK CHECK

1. What is misinformation?

A A joke website
B False information made to deceive
C False information shared by someone who believes it
D True information shared to harm

2. What is disinformation?

A An advert
B False information created or shared on purpose to deceive
C A spelling mistake
D A true story

3. Someone posts a classmate's real private messages to humiliate them. This is...

A malinformation
B disinformation
C satire
D misinformation

4. Why do some websites make up shocking stories?

A To improve the weather
B Clicks can earn advertising money
C It is required by law
D They are bored

QUICK CHECK: ANSWERS

1. **C: False information shared by someone who believes it**
   Misinformation is shared by mistake.

2. **B: False information created or shared on purpose to deceive**
   Disinformation is deliberate.

3. **A: malinformation**
   True information, shared to harm.

4. **B: Clicks can earn advertising money**
   Money is a common motive.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Define misinformation, disinformation and malinformation.

Look back at today's success criteria: can you tick them all?

- ✓ I can define misinformation, disinformation and malinformation
- ✓ I can sort real examples into the three types
- ✓ I can explain why intent matters

LESSON 2 OF 6

HOW DEEPFAKES ARE MADE

Deepfakes are realistic fake media made with AI: face swaps, lip-syncing, voice clones and fully generated video.

STARTER · 5 MINUTES

DO NOW

How many photos of your face do you think are online right now? Who could use them?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how AI is used to create deepfake images, voices and videos

SUCCESS CRITERIA: I CAN...

- ✓ I can describe four types of deepfake
- ✓ I can explain how AI learns to copy a face or voice
- ✓ I can explain why deepfakes are getting harder to spot

WHAT IS A DEEPFAKE?

- Fake image, audio or video made with AI that looks or sounds like a real person
- The name comes from "deep learning" + "fake"
- Can show someone doing or saying something they never did
- Tools are now free, fast and run on a phone

KEY WORDS

DEEPFAKE	An image, audio clip or video made or changed with AI so that a real person seems to do or say something they never did.
FACE SWAP	A deepfake that places one person's face onto someone else's body.
VOICE CLONE	An AI copy of a person's voice, made by training on recordings of them.
GENERATIVE AI	Artificial intelligence that creates new content such as text, images, audio or video.

FOUR TYPES

- Face swap: one person's face placed onto another's body
- Lip-sync: real video, but the mouth is changed to match new words
- Voice clone: a copy of someone's voice from a few seconds of audio
- Fully generated: a whole video or photo created from a text prompt

ACTIVITY · 15 MINUTES · PAIRS

DEEPFAKE DETECTIVES

1. Your teacher shows six short clips or images (some real, some AI-made, all of fictional or consenting people).
2. For each one, decide real or fake and write the clue you used.
3. Your teacher reveals the answers: how many did you get right?

In your workbook, answer:

? How many did you get right, and which clue was most useful?

? Why might checking the source be more reliable than looking for glitches?

Your workbook: Lesson 2 · tick each step off as you go

HOW DOES THE AI LEARN?

- It studies many examples of a face or voice
- It learns patterns: how the face moves, how the voice sounds
- It then generates new frames or sounds that fit those patterns
- More examples = more convincing fake

QUICK CHECK

1. What is a deepfake?

A A video game
B A blurry video
C A very old photo
D AI-made media showing a real person doing or saying something they didn't

2. What does a voice clone do?

A Blocks phone calls
B Turns speech into text
C Copies someone's voice from recordings
D Makes your voice louder

3. Why are celebrities easy targets for deepfakes?

A Their faces are simpler
B AI only knows famous people
C They ask for deepfakes
D There is lots of video and audio of them online to train AI

4. Why is spotting deepfakes by eye becoming less reliable?

A Screens are worse
B Deepfakes are banned
C AI tools keep improving and remove obvious glitches
D People's eyesight is worse

QUICK CHECK: ANSWERS

1. **D: AI-made media showing a real person doing or saying something they didn't**
   Deepfakes fake real people with AI.

2. **C: Copies someone's voice from recordings**
   Voice cloning imitates a real voice.

3. **D: There is lots of video and audio of them online to train AI**
   More examples make better fakes.

4. **C: AI tools keep improving and remove obvious glitches**
   Better tools mean fewer visible errors.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how an AI system is able to create a voice clone of a person.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe four types of deepfake
- ✓ I can explain how AI learns to copy a face or voice
- ✓ I can explain why deepfakes are getting harder to spot

LESSON 3 OF 6

FACT-CHECKER'S TOOLKIT

Professional fact-checkers use tools as well as their eyes.

STARTER · 5 MINUTES

DO NOW

A photo shows a shark swimming down a flooded motorway. How could you find out if it's real?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Use reverse image search, metadata and content credentials to investigate images

SUCCESS CRITERIA: I CAN...

- ✓ I can use a reverse image search to find a picture's origin
- ✓ I can explain what metadata is and what it can show
- ✓ I can explain what content credentials are

WHERE DID IT COME FROM?

- Reverse image search: search with a picture instead of words
- Shows where else the image appears online, and often the original
- Old photos are often reshared as if they were new
- Try Google Lens, Bing Visual Search or TinEye

KEY WORDS

REVERSE IMAGE SEARCH	Searching the internet using a picture instead of words, to find where else it appears.
METADATA	Hidden information stored with a file, such as the date, device and location of a photo.
CONTENT CREDENTIALS	A secure record attached to a file showing who made it, with what tool and how it was edited.
PROVENANCE	The history of where something came from and how it has changed.

DATA ABOUT DATA

- Metadata is hidden information stored with a file
- Photos can hold the date, the camera or phone, settings and sometimes location
- Missing or odd metadata can be a clue, but isn't proof
- Social media usually strips metadata when you upload

A HISTORY LABEL FOR MEDIA

- Provenance means the history of where something came from
- Content credentials are a new standard (C2PA) that records how a file was made and edited
- Some cameras and AI tools add them automatically
- Look for the "CR" icon or check with a content credentials inspector

ACTIVITY · 25 MINUTES · PAIRS

INVESTIGATE THE VIRAL IMAGE

1. Your teacher gives you three images that "went viral".

2. For each, do a reverse image search and note the earliest date and original source you can find.

3. Look for any metadata or content credentials (your teacher may provide the files).

4. Write a verdict: real, real but old/misleading, edited, or AI-generated, and your evidence.

In your workbook, answer:

? Write your verdict and evidence for one image.

? Which tool was most useful, and why?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What does a reverse image search do?

A Searches the web using a picture to find where it appears
B Makes the image bigger
C Flips the image
D Deletes the image

2. What is metadata?

A The main picture
B Hidden information stored with a file, such as date and device
C A type of virus
D The file's colours

3. What are content credentials?

A A secure record of how a file was made and edited
B A filter on a camera
C A photographer's degree
D A password for a website

4. A "new" storm photo appears in search results from years ago. What does this suggest?

A Search engines are broken
B The storm happened twice
C It's definitely AI
D It's an old photo being reshared as new

QUICK CHECK: ANSWERS

1. **A: Searches the web using a picture to find where it appears**
   You search with an image instead of words.

2. **B: Hidden information stored with a file, such as date and device**
   Metadata is data about data.

3. **A: A secure record of how a file was made and edited**
   They record a file's history, including AI use.

4. **D: It's an old photo being reshared as new**
   Recycled photos are a common trick.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a reverse image search can help show that a photo is misleading.

Look back at today's success criteria: can you tick them all?

- ✓ I can use a reverse image search to find a picture's origin
- ✓ I can explain what metadata is and what it can show
- ✓ I can explain what content credentials are

LESSON 4 OF 6

THINK LIKE A FACT-CHECKER

Professional fact-checkers don't just study a website: they leave it and see what other trusted sources say about it.

STARTER · 5 MINUTES

DO NOW

A website looks really professional and says it's "independent". Does that mean you can trust it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Check claims using lateral reading and the SIFT method

SUCCESS CRITERIA: I CAN...

- ✓ I can use the four steps of SIFT
- ✓ I can read laterally to check a source
- ✓ I can judge whether a source is reliable

LEAVE THE PAGE

- Looking professional doesn't make a site reliable
- Fact-checkers read laterally: they open new tabs to see what others say about the source
- Students who read "vertically" (staying on the page) are much easier to fool
- Lateral reading is quicker and more accurate

KEY WORDS

LATERAL READING	Checking a source by leaving it and seeing what other trusted sources say about it.
SIFT	A four-step method for checking information: Stop, Investigate the source, Find better coverage, Trace to the original.
SOURCE	Where a piece of information comes from.

SIFT

- Stop: notice your emotions and don't share yet
- Investigate the source: who made this and why?
- Find better coverage: do trusted outlets report it?
- Trace claims, quotes and media back to the original

ACTIVITY · 20 MINUTES · PAIRS

SIFT RACE

1. Your teacher gives each pair four claims found online.
2. Use the SIFT steps to check each claim. Time yourselves.
3. Record what you found at each step and your verdict.

In your workbook, answer:

? Choose one claim. Write what you found for each SIFT step.

? Why is "Stop" the first step?

Your workbook: Lesson 4 · tick each step off as you go

IS THIS SOURCE RELIABLE?

- Who is behind it, and do they have expertise?
- Does it have a reason to mislead (money, politics)?
- Do other reliable sources say the same?
- Is it up to date, and does it link to evidence?

QUICK CHECK

1. What is lateral reading?

A Reading a page from top to bottom
B Leaving a page to check what other sources say about it
C Skimming the headlines only
D Reading sideways on a phone

2. What does the S in SIFT stand for first?

A Scroll
B Share
C Search
D Stop

3. What does "Trace claims to the original" mean?

A Copy the claim
B Delete the claim
C Find where a quote, statistic or picture really came from
D Draw over the picture

4. Which is a sign a source may be unreliable?

A Links to evidence
B No named authors and a history of clickbait
C Named experts
D Corrections when it makes mistakes

QUICK CHECK: ANSWERS

1. **B: Leaving a page to check what other sources say about it**
   Fact-checkers read across sources.

2. **D: Stop**
   Stop before you react or share.

3. **C: Find where a quote, statistic or picture really came from**
   Tracing reveals missing context.

4. **B: No named authors and a history of clickbait**
   Anonymous clickbait sites are risky.

4 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe the four steps of the SIFT method.

Look back at today's success criteria: can you tick them all?

- ✓ I can use the four steps of SIFT
- ✓ I can read laterally to check a source
- ✓ I can judge whether a source is reliable

LESSON 5 OF 6

WHY LIES TRAVEL FAST

False stories often spread faster than true ones because they are surprising and emotional.

DO NOW

Which post would you be more likely to share: "Local council meeting goes well" or "SHOCKING: what they're hiding at the council"? Why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why false information spreads quickly online

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how emotion makes content spread
- ✓ I can explain how algorithms and bots amplify content
- ✓ I can describe echo chambers and filter bubbles

EMOTION IS THE ENGINE

- Content that causes shock, anger or fear gets shared most
- False stories are often more surprising than true ones
- A famous study found false news reached people much faster than true news on Twitter
- Sharing takes one tap; checking takes effort

KEY WORDS

ALGORITHM	A set of rules a computer follows; on social media, the rules that choose what appears in your feed.
ENGAGEMENT	How people interact with content: likes, comments, shares and watch time.
BOT	An automated account run by software rather than a person.
ECHO CHAMBER	A space where you mostly hear from people who already agree with you.
FILTER BUBBLE	When an algorithm keeps showing you content similar to what you already like, so you see fewer other views.

MACHINES THAT AMPLIFY

- Platform algorithms choose what you see, often by engagement: likes, comments, shares, watch time
- Outrage gets engagement, so it can get promoted
- Bots: automated accounts that post and share at huge scale
- Coordinated fake accounts can make a few people look like a crowd

BUBBLES AND CHAMBERS

- Filter bubble: the algorithm shows you more of what you already like
- Echo chamber: you mostly hear from people who agree with you
- Inside a bubble, false claims go unchallenged
- Following a wide range of trusted sources helps you see more of the picture

ACTIVITY · 20 MINUTES · GROUPS OF 3

TRACE THE SPREAD

1. Your teacher shows how a made-up story about a local school spread over 24 hours (a simulation with fictional accounts).

2. Mark on the timeline where the story got a boost: an emotional headline, a bot network, an influencer share, the algorithm.

3. Mark the three points where someone could have stopped it.

In your workbook, answer:

? Name two things that helped the story spread and explain how.

? Where could the spread have been stopped, and by whom?

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. Why do emotional posts spread quickly?

A People share them before thinking or checking
B Algorithms hide them
C They are shorter
D They are always true

2. What is engagement on social media?

A Turning off notifications
B Likes, comments, shares and watch time
C The number of followers you block
D Getting married

3. What is a bot?

A An automated account run by software
B A type of camera
C A person who checks facts
D A news website

4. What is a filter bubble?

A A bubble filter for water
B A privacy setting
C A type of virus
D When the algorithm keeps showing you what you already like

QUICK CHECK: ANSWERS

1. **A: People share them before thinking or checking**
   Emotion drives quick sharing.

2. **B: Likes, comments, shares and watch time**
   Feeds often reward engagement.

3. **A: An automated account run by software**
   Bots post and share automatically.

4. **D: When the algorithm keeps showing you what you already like**
   It narrows what you see.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a social media algorithm can help false information spread.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how emotion makes content spread
- ✓ I can explain how algorithms and bots amplify content
- ✓ I can describe echo chambers and filter bubbles

LESSON 6 OF 6

FIGHTING BACK

Laws, platform rules, fact-checkers and new technology all help, but individuals matter too.

STARTER · 5 MINUTES

DO NOW

If you saw a friend share something false, would you tell them? How would you do it without starting an argument?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how the law, platforms and individuals can reduce the harm of deepfakes and misinformation

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how laws and platforms tackle harmful fakes
- ✓ I can explain what prebunking is
- ✓ I can design a campaign to protect others

RULES AND LAWS

- The UK's Online Safety Act makes big platforms deal with illegal and harmful content
- Sharing intimate images of someone without consent, including deepfakes, is a crime in the UK
- Platforms have rules: removing, labelling or slowing false and harmful posts
- Elections have rules about who pays for political adverts

KEY WORDS

PREBUNKING	Learning about manipulation tricks before you encounter them, so you can recognise them.
FACT-CHECKER	A person or organisation that investigates claims and publishes what is true.
ONLINE SAFETY ACT	A UK law that requires online platforms to protect users from illegal content and children from harmful content.

HELPERS

- Professional fact-checkers: Full Fact, BBC Verify, FactCheckNI
- Content credentials and AI labels (lesson 3)
- Community notes: users add context to misleading posts
- Teachers, parents and trusted adults

PREBUNKING AND YOU

- Prebunking: learning the tricks before you meet them, like a vaccine
- Spot the tricks: fake experts, emotional language, out-of-context pictures, impersonation
- Correct kindly and privately, with a good source
- Report harmful fakes; never pass them on

ACTIVITY · 30 MINUTES · GROUPS OF 4

PREBUNK CAMPAIGN

1. Choose one manipulation trick from the list.
2. Design a short prebunking campaign for Year 7: a poster, a 30-second video script or a set of three social media posts.
3. Show the trick with a made-up, harmless example (no real people), and give one tip for spotting it.
4. Present to another group and give each other feedback.

In your workbook, answer:

? Which trick did you choose, and what is your one-sentence tip for spotting it?

? Write a kind private message you could send to a friend who shared a false post.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What does the Online Safety Act require big platforms to do?

A Protect users from illegal content and children from harmful content
B Pay every user
C Delete all posts
D Stop people using phones

2. What is prebunking?

A Correcting a post after it spreads
B Fact-checking a picture
C Learning about manipulation tricks before you meet them
D Deleting your account

3. What is the best way to correct a friend who shared something false?

A Report them to the police
B Ignore it and share it too
C Mock them in the comments
D Message them privately and kindly with a reliable source

4. Which organisation fact-checks claims in Northern Ireland?

A NI Water
B Translink
C FactCheckNI
D The Met Office

QUICK CHECK: ANSWERS

1. **A: Protect users from illegal content and children from harmful content**
   It sets safety duties for platforms.

2. **C: Learning about manipulation tricks before you meet them**
   Prebunking works like a vaccine.

3. **D: Message them privately and kindly with a reliable source**
   Kindness and evidence work best.

4. **C: FactCheckNI**
   FactCheckNI checks claims in NI.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three things a pupil should do if a fake image of them is shared online.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how laws and platforms tackle harmful fakes
- ✓ I can explain what prebunking is
- ✓ I can design a campaign to protect others

END OF TOPIC

DEEPFAKES AND MISINFORMATION

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.